作业5

本后题：

e.x. 8.6

(2). $\because \lim\limits_{x\to 0}(2+x)=2$，令 $xy=u$.

$$\lim_{(x,y)\to(0,0)}\frac{(2+x)\ln(1+xy)}{xy}=\lim_{x\to 0}(2+x)\lim_{u\to 0}\frac{\ln(1+u)}{u}=2.$$

(4). 令 $\sqrt{x^2+y^2}=u$

$\therefore$ 原式 $=\lim\limits_{u\to 0} u\sin\frac{1}{u}$

$=\lim\limits_{u\to 0}\frac{\sin\frac{1}{u}}{\frac{1}{u}}$ 令 $\frac{1}{u}=v$ $\therefore$ 上式 $=\lim\limits_{v\to\infty}\frac{\sin v}{v}=0.$

7. (2). 令 $y=kx$

$\lim\limits_{(x,y)\to(0,0)} f(x,y)=\lim\limits_{x\to 0}\frac{kx^2}{(1+k)x}$ 令 $k=-1$，此时极限不存在.

$\therefore$ 极限不存在.

8. (2). $\sin\pi x=0$ or $\sin\pi y=0$.

$\therefore$ 在直线 $x=m$ 和 $y=n$ 上

(4). 在 $x=0$ 或 $y=0$ 或 $x\delta=0$ 时

9. (2). 原式 $=\left.\frac{1-\frac{y}{2x^2}}{x+\frac{y}{2x}}\right|_{(1,0)}=1$

(4). $f_x(x,1) = 1 + (y-1)\cdot \dfrac{\sqrt{\frac{1}{y}}\cdot\frac{1}{2\sqrt{x}}}{\sqrt{1-\frac{x}{y}}} = 1$

10. (2). $\dfrac{\partial z}{\partial x} = -\left(\frac{1}{3}\right)^{-\frac{y}{x}} \ln 3 \cdot \dfrac{y}{x^2}$

$\dfrac{\partial z}{\partial y} = \left(\frac{1}{3}\right)^{-\frac{y}{x}} \ln 3 \cdot \dfrac{1}{x}$

(4). $\dfrac{\partial z}{\partial x} = \dfrac{ye^{xy}(e^x+e^y) - e^{xy}(e^x)}{(e^x+e^y)^2} = \dfrac{e^{xy}(ye^x+ye^y-e^x)}{(e^x+e^y)^2}$.

$\dfrac{\partial z}{\partial y} = \dfrac{e^{xy}(xe^y+xe^x-e^y)}{(e^x+e^y)^2}$.

(6). $\dfrac{\partial z}{\partial x} = \dfrac{-2y}{\sqrt{1-(3-2xy)^2}} - \cos\left(3-\frac{2x}{y}\right)\cdot\dfrac{2}{y}$.

$\dfrac{\partial z}{\partial y} = \dfrac{-2x}{\sqrt{1-(3-2xy)^2}} + \cos\left(3-\frac{2x}{y}\right)\cdot\left(\frac{2x}{y^2}\right)$

x x

(8). $\ln z = (x+y)\ln(1+xy)$

$\dfrac{1}{z}\dfrac{\partial z}{\partial x} = \ln(1+xy) + (x+y)\cdot\dfrac{y}{1+xy}$

$\dfrac{\partial z}{\partial x} = (1+xy)^{x+y}\left(\ln(1+xy) + (x+y)\cdot\dfrac{y}{1+xy}\right)$

$\dfrac{\partial z}{\partial y} = (1+xy)^{x+y}\left(\ln(1+xy) + (x+y)\cdot\dfrac{x}{1+xy}\right)$

(10). $\frac{\partial u}{\partial x} = e^{x(x^2+y^2+z^2)} \cdot (3x^2+y^2+z^2)$

$\frac{\partial u}{\partial y} = e^{x(x^2+y^2+z^2)} \cdot (2xy)$

$\frac{\partial u}{\partial z} = e^{x(x^2+y^2+z^2)} \cdot (2xz)$

11. (2). $\left.\frac{\partial z}{\partial x}\right|_{x=1} = \left.\frac{\partial z}{\partial y}\right|_{y=1} = \frac{2}{2\sqrt{1+1+1}} = \frac{1}{\sqrt{3}}$

$\because x=1$ $\therefore z=\sqrt{2+y^2}$. $\therefore$ 切向量 $(0,1,\frac{1}{\sqrt{3}})$，并 $(1,1,\sqrt{3})$ 代入

$\therefore$ 直线参数方程为 $\begin{cases} x=1 \\ y=1+t \\ z=\frac{1}{\sqrt{3}}t+\sqrt{3} \end{cases}$

$\therefore$ 平面法向量 $\vec{n}=(0,1,\frac{1}{\sqrt{3}})$

$\therefore$ 平面方程为

$(y-1)+\frac{1}{\sqrt{3}}(z-\sqrt{3})=0$ 即 $\sqrt{3}y+z-2\sqrt{3}=0$.

12. (2) $\frac{\partial z}{\partial x} = \frac{1}{1+(\frac{x+y}{1-xy})^2} \cdot \left(\frac{1-xy+y(x+y)}{(1-xy)^2}\right)$

$= \frac{1}{x^2+1}$

$\therefore \frac{\partial^2 z}{\partial x^2} = -\frac{2x}{(x^2+1)^2}$, $\frac{\partial^2 z}{\partial x \partial y} = 0$.

$\frac{\partial^2 z}{\partial y^2} = -\frac{2y}{(y^2+1)^2}$

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(4). $\frac{\partial z}{\partial x} = y^{\ln x} \cdot \ln y \cdot \frac{1}{x}$

$$\frac{\partial^2 z}{\partial x^2} = \ln y \cdot \frac{y^{\ln x} \ln y - y^{\ln x}}{x^2} = y^{\ln x} \ln y \frac{\ln y - 1}{x^2}$$

$$\frac{\partial^2 z}{\partial x \partial y} = \left(\ln x \cdot y^{\ln x - 1} \ln y + y^{\ln x} \cdot \frac{1}{y}\right) \cdot \frac{1}{x}$$

$$= \frac{1}{x}(\ln x \ln y + 1) \cdot y^{\ln x - 1}$$

$$\frac{\partial^2 z}{\partial y} = \ln x \cdot y^{\ln x - 1}$$

$$\frac{\partial^2 z}{\partial y^2} = \ln x (\ln x - 1) y^{\ln x - 2}.$$

13. (2). $\frac{\partial z}{\partial x} = \frac{1}{\sqrt{x^2+y^2}} \cdot \frac{2x}{2\sqrt{x^2+y^2}} = \frac{x}{x^2+y^2}$

$$\frac{\partial^2 z}{\partial x^2} = \frac{(x^2+y^2) - x \cdot 2x}{(x^2+y^2)^2} = \frac{y^2 - x^2}{(x^2+y^2)^2}.$$

$$\therefore \frac{\partial^2 z}{\partial y^2} = \frac{x^2 - y^2}{(x^2+y^2)^2}, \quad \therefore \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0.$$

14. (2). $\frac{\partial u}{\partial x} = \frac{1}{1+\left(\frac{y}{x}\right)^2} \cdot \left(-\frac{y}{x^2}\right) + \frac{1}{1+\left(\frac{z}{x}\right)^2} \cdot \left(-\frac{z}{x^2}\right)$

$$= -\frac{y}{x^2+y^2} - \frac{z}{x^2+z^2}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{2xy}{(x^2+y^2)^2} + \frac{2xz}{(x^2+z^2)^2}$$

$$\frac{\partial u}{\partial y} = \frac{\frac{1}{x}}{1+\left(\frac{y}{x}\right)^2} = \frac{x}{x^2+y^2}, \quad \frac{\partial^2 u}{\partial y^2} = -\frac{2xy}{(x^2+y^2)}$$

$$\frac{\partial^2 u}{\partial z^2} = -\frac{2xz}{(x^2+z^2)} \quad \therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0.$$

(4). $$\frac{\partial(\ln r)}{\partial x} = \frac{1}{r} \cdot \frac{2x}{2\sqrt{x^2+y^2+z^2}} = \frac{x}{x^2+y^2+z^2}.$$

$$\frac{\partial^2(\ln r)}{\partial x^2} = \frac{(x^2+y^2+z^2) - 2x^2}{(x^2+y^2+z^2)^2} = \frac{-x^2+y^2+z^2}{(x^2+y^2+z^2)^2}.$$

$$\frac{\partial^2(\ln r)}{\partial y^2} = \frac{x^2-y^2+z^2}{(x^2+y^2+z^2)^2}, \quad \frac{\partial^2(\ln r)}{\partial z^2} = \frac{x^2+y^2-z^2}{(x^2+y^2+z^2)^2}$$

$$\therefore \frac{\partial^2(\ln r)}{\partial x^2} + \frac{\partial^2(\ln r)}{\partial y^2} + \frac{\partial^2(\ln r)}{\partial z^2} = \frac{x^2+y^2+z^2}{(x^2+y^2+z^2)^2} = \frac{1}{x^2+y^2+z^2} = \frac{1}{r^2}$$

16.(2). $$\frac{\partial z}{\partial x} = \frac{\sqrt{x^2+y^2} - \frac{x \cdot 2x}{2\sqrt{x^2+y^2}}}{x^2+y^2} = \frac{y^2}{(x^2+y^2)^{\frac{3}{2}}}$$

$$\frac{\partial z}{\partial y} = -\frac{1}{2} x (x^2+y^2)^{-\frac{3}{2}} \cdot 2y = -xy(x^2+y^2)^{-\frac{3}{2}}$$

$$\therefore dz = y^2 (x^2+y^2)^{-\frac{3}{2}} dx - xy(x^2+y^2)^{-\frac{3}{2}} dy.$$

(4). $$\frac{\partial u}{\partial x} = \frac{1}{yz}\left(\frac{x}{y}\right)^{\frac{1}{z}-1}, \quad \frac{\partial u}{\partial y} = -\frac{x}{y^2} \cdot \frac{1}{z}\left(\frac{x}{y}\right)^{\frac{1}{z}-1}, \quad \frac{\partial u}{\partial z} = -\frac{1}{z^2} \cdot \left(\frac{x}{y}\right)^{\frac{1}{z}} \ln\left(\frac{x}{y}\right)$$

$$\therefore du = \frac{1}{yz}\left(\frac{x}{y}\right)^{\frac{1}{z}-1} dx - \frac{x}{zy^2}\left(\frac{x}{y}\right)^{\frac{1}{z}-1} dy - \frac{1}{z^2}\left(\frac{x}{y}\right)^{\frac{1}{z}} \ln\frac{x}{y}\, dz.$$

思考题:

一. U是S和V的函数

F是T和V的函数

H是S和P的函数

G是T和P的函数

$$\frac{\partial U}{\partial S}=T,\ \frac{\partial U}{\partial V}=-P;\ \frac{\partial F}{\partial T}=-S,\ \frac{\partial F}{\partial V}=-P$$

$$\frac{\partial H}{\partial S}=T,\ \frac{\partial S}{\partial P}=V;\ \frac{\partial G}{\partial T}=-S,\ \frac{\partial G}{\partial P}=V$$

二.

1). $dU=\left(\frac{\partial U}{\partial S}\right)_V dS+\left(\frac{\partial U}{\partial V}\right)_S dV$

$\therefore \left(\frac{\partial U}{\partial S}\right)_V=T;\ \left(\frac{\partial U}{\partial V}\right)_S=-P$

2) $\left(\frac{\partial U}{\partial x}\right)_y=\frac{\partial U}{\partial S}\frac{\partial S}{\partial x}+\frac{\partial U}{\partial V}\frac{\partial V}{\partial x}=T\left(\frac{\partial S}{\partial x}\right)_y-P\left(\frac{\partial V}{\partial x}\right)_y$

$\left(\frac{\partial U}{\partial y}\right)_x=\frac{\partial U}{\partial S}\frac{\partial S}{\partial y}+\frac{\partial U}{\partial V}\frac{\partial V}{\partial y}=T\left(\frac{\partial S}{\partial y}\right)_x-P\left(\frac{\partial V}{\partial y}\right)_x$

3). $\because \frac{\partial}{\partial y}\left(\frac{\partial U}{\partial x}\right)=\frac{\partial}{\partial x}\left(\frac{\partial U}{\partial y}\right)$

$\frac{\partial}{\partial y}\left(\frac{\partial U}{\partial x}\right)=\frac{\partial T}{\partial y}\left(\frac{\partial S}{\partial x}\right)+T\frac{\partial^2 S}{\partial x\partial y}-\frac{\partial P}{\partial y}\left(\frac{\partial V}{\partial x}\right)-P\frac{\partial^2 V}{\partial x\partial y}$

$\frac{\partial}{\partial x}\left(\frac{\partial U}{\partial y}\right)=\frac{\partial T}{\partial x}\frac{\partial S}{\partial y}+T\frac{\partial^2 S}{\partial x\partial y}-\frac{\partial P}{\partial x}\frac{\partial V}{\partial y}-P\frac{\partial^2 V}{\partial x\partial y}$

$$\therefore \left(\frac{\partial T}{\partial y}\right)_x\left(\frac{\partial S}{\partial x}\right)_y-\left(\frac{\partial P}{\partial y}\right)_x\left(\frac{\partial V}{\partial x}\right)_y=\left(\frac{\partial T}{\partial x}\right)_y\left(\frac{\partial S}{\partial y}\right)_x-\left(\frac{\partial P}{\partial x}\right)_y\left(\frac{\partial V}{\partial y}\right)_x$$

4). 令 $x=S$, $y=V$

$$\therefore \left(\frac{\partial T}{\partial V}\right)_S\times 1-\left(\frac{\partial P}{\partial V}\right)_S\times\left(\frac{\partial V}{\partial S}\right)_V=\left(\frac{\partial T}{\partial S}\right)_V\left(\frac{\partial S}{\partial V}\right)_S-\left(\frac{\partial P}{\partial S}\right)_V\times 1$$

$$\because \left(\frac{\partial V}{\partial S}\right)_V=\left(\frac{\partial S}{\partial V}\right)_S=0.$$

$$\therefore \left(\frac{\partial T}{\partial V}\right)_S=-\left(\frac{\partial P}{\partial S}\right)_V$$ 这是Maxwell关系式之一

5). $dH=TdS+VdP$

$$\therefore \frac{\partial H}{\partial S}=T,\ \frac{\partial H}{\partial P}=V$$

$$\therefore \frac{\partial}{\partial P}\left(\frac{\partial H}{\partial S}\right)=\frac{\partial}{\partial S}\left(\frac{\partial H}{\partial P}\right)$$

$$\left(\frac{\partial T}{\partial P}\right)_S=\left(\frac{\partial V}{\partial S}\right)_P \quad (*)$$

此已是(3)中最终表达式中令 $x=S$, $y=P$ 的结果.

亦令 $S=x$, $y=P$

$$\therefore \left(\frac{\partial T}{\partial y}\right)_x\left(\frac{\partial S}{\partial x}\right)_y=\left(\frac{\partial T}{\partial P}\right)_S;\ \left(\frac{\partial P}{\partial y}\right)_x\left(\frac{\partial V}{\partial x}\right)_y=\left(\frac{\partial V}{\partial S}\right)_y.$$

$$\left(\frac{\partial T}{\partial x}\right)_y\left(\frac{\partial S}{\partial y}\right)_x=\left(\frac{\partial P}{\partial x}\right)_y\left(\frac{\partial V}{\partial y}\right)_x=0.$$ 由(*)有

$$\therefore \left(\frac{\partial T}{\partial y}\right)_x\left(\frac{\partial S}{\partial x}\right)_y-\left(\frac{\partial P}{\partial y}\right)_x\left(\frac{\partial V}{\partial x}\right)_y=\left(\frac{\partial T}{\partial x}\right)_y\left(\frac{\partial S}{\partial y}\right)_x-\left(\frac{\partial P}{\partial x}\right)_y\left(\frac{\partial V}{\partial y}\right)_x$$

此即(3)中方程.

6) 对态函数 U ：对任一闭合路径

$$\oint dU = 0.$$

在 S-V 平面上：取以 (S, V) 和 $(S+\Delta S, V+\Delta V)$ 为对角线 $(S+\Delta S, V)$；$(S, V+\Delta V)$ 为另一对角线的小矩形

考虑矩形上的路径积分.

ΔU_1: $(S, V) \to (S+\Delta S, V)$

$\Delta U_1 = T(S, V)\Delta S$

ΔU_2: $(S+\Delta S, V) \to (S+\Delta S, V+\Delta V)$

$\Delta U_2 = P(S+\Delta S, V)\Delta V$

ΔU_3: $(S+\Delta S, V+\Delta V) \to (S, V+\Delta V)$

$\Delta U_3 = -T(S+\Delta S, V+\Delta V)\Delta S$

ΔU_4: $(S, V+\Delta V) \to (S, V)$

$\Delta U_4 = -P(S, V+\Delta V)\Delta V$

$\because \sum \Delta U_i = 0$，二阶及以上小量略去，有

$P(S+\Delta S, V) \approx P(S, V) + \left(\frac{\partial P}{\partial S}\right)_V \Delta S$

$T(S+\Delta S, V+\Delta V) \approx T(S, V) + \left(\frac{\partial T}{\partial S}\right)_V \Delta S + \left(\frac{\partial T}{\partial V}\right)_S \Delta V$

$P(S, V+\Delta V) = P(S, V) + \left(\frac{\partial P}{\partial V}\right)_S \Delta V$ 代入 $\sum \Delta U_i = 0$ 有

$\therefore T(S, V)\Delta S + \left(P(S, V) + \left(\frac{\partial P}{\partial S}\right)_V \Delta S\right)\Delta V - \left(T(S, V) + \left(\frac{\partial T}{\partial S}\right)_V \Delta S + \left(\frac{\partial T}{\partial V}\right)_S \Delta V\right)\Delta S - \left(P(S, V) + \left(\frac{\partial P}{\partial V}\right)_S \Delta V\right)\Delta V = 0$

$\therefore \left(\left(\frac{\partial P}{\partial S}\right)_V + \left(\frac{\partial T}{\partial V}\right)_S\right)\Delta S \Delta V = 0$

$\because \Delta S, \Delta V$ 均是任取的

$\therefore \left(\frac{\partial P}{\partial S}\right)_V + \left(\frac{\partial T}{\partial V}\right)_S = 0$ 即 $\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$

此即为(3)中所得 Maxwell 关系.